

JEE Mains 2019 Chapter wise Question Bank

Inverse Trigonometric Functions – Questions

Q1

If $\cos^{-1}\left(\frac{2}{3x}\right) + \cos^{-1}\left(\frac{3}{4x}\right) = \frac{\pi}{2}$ ($x > \frac{3}{4}$),

then x is equal to:

- (1) $\frac{\sqrt{145}}{12}$ (2) $\frac{\sqrt{145}}{10}$
 (3) $\frac{\sqrt{146}}{12}$ (4) $\frac{\sqrt{145}}{11}$

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Q2

If $x = \sin^{-1}(\sin 10)$ and $y = \cos^{-1}(\cos 10)$, then $y - x$ is equal to:

- (1) 0 (2) 10
 (3) 7π (4) π

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Q3

The value of $\cot\left(\sum_{n=1}^{19} \cot^{-1}\left(1 + \sum_{p=1}^n 2p\right)\right)$ is:

- (1) $\frac{21}{19}$ (2) $\frac{19}{21}$
 (3) $\frac{22}{23}$ (4) $\frac{23}{22}$

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Q4

All x satisfying the inequality $(\cot^{-1}x)^2 - 7(\cot^{-1}x) + 10 > 0$, lie in the interval :

- (1) $(-\infty, \cot 5) \cup (\cot 4, \cot 2)$
 (2) $(\cot 2, \infty)$
 (3) $(-\infty, \cot 5) \cup (\cot 2, \infty)$
 (4) $(\cot 5, \cot 4)$

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Q5

Considering only the principal values of inverse functions,

the set $A = \left\{x \geq 0 : \tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}\right\}$

- (1) contains two elements
 (2) contains more than two elements
 (3) is a singleton
 (4) is an empty set

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Q6

If $\alpha = \cos^{-1}\left(\frac{3}{5}\right)$, $\beta = \tan^{-1}\left(\frac{1}{3}\right)$, where $0 < \alpha, \beta < \frac{\pi}{2}$, then

$\alpha - \beta$ is equal to :

- (1) $\tan^{-1}\left(\frac{9}{5\sqrt{10}}\right)$ (2) $\cos^{-1}\left(\frac{9}{5\sqrt{10}}\right)$
 (3) $\tan^{-1}\left(\frac{9}{14}\right)$ (4) $\sin^{-1}\left(\frac{9}{5\sqrt{10}}\right)$

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Q7

If $\cos^{-1} x - \cos^{-1} \frac{y}{2} = \alpha$, where $-1 \leq x \leq 1$, $-2 \leq y \leq 2$,

$x \leq \frac{y}{2}$, then for all x, y , $4x^2 - 4xy \cos \alpha + y^2$ is equal to:

- (1) $4 \sin^2 \alpha$ (2) $2 \sin^2 \alpha$
(3) $4 \sin^2 \alpha - 2x^2 y^2$ (d) $4 \cos^2 \alpha + 2x^2 y^2$

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Q8

The value of $\sin^{-1} \left(\frac{12}{13} \right) - \sin^{-1} \left(\frac{3}{5} \right)$ is equal to :

- (1) $\pi - \sin^{-1} \left(\frac{63}{65} \right)$ (2) $\frac{\pi}{2} - \sin^{-1} \left(\frac{56}{65} \right)$
(3) $\frac{\pi}{2} - \cos^{-1} \left(\frac{9}{65} \right)$ (4) $\pi - \cos^{-1} \left(\frac{33}{65} \right)$

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Inverse Trigonometric Functions - Solutions

Q1

$$(1) \cos^{-1}\left(\frac{2}{3x}\right) + \cos^{-1}\left(\frac{3}{4x}\right) = \frac{\pi}{2}; \left(x > \frac{3}{4}\right)$$

$$\Rightarrow \cos^{-1}\left(\frac{2}{3x}\right) = \frac{\pi}{2} - \cos^{-1}\left(\frac{3}{4x}\right)$$

$$\Rightarrow \cos^{-1}\left(\frac{2}{3x}\right) = \sin^{-1}\left(\frac{3}{4x}\right)$$

$$\left[\because \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \right]$$

$$\text{Put } \sin^{-1}\left(\frac{3}{4x}\right) = \theta \Rightarrow \sin \theta = \frac{3}{4x}$$

$$\Rightarrow \cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \frac{9}{16x^2}}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{\sqrt{16x^2 - 9}}{4x}\right)$$

$$\therefore \cos^{-1}\left(\frac{2}{3x}\right) = \cos^{-1}\left(\frac{\sqrt{16x^2 - 9}}{4x}\right)$$

$$\Rightarrow \frac{2}{3x} = \frac{\sqrt{16x^2 - 9}}{4x} \Rightarrow x^2 = \frac{64 + 81}{9 \times 16} \Rightarrow x = \pm \sqrt{\frac{145}{144}}$$

$$\Rightarrow x = \frac{\sqrt{145}}{12} \quad \left(\because x > \frac{3}{4} \right)$$

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Q2

$$(4) x = \sin^{-1}(\sin 10)$$

$$\Rightarrow x = 3\pi - 10 \quad \begin{cases} 3\pi - \frac{\pi}{2} < 10 < 3\pi + \frac{\pi}{2} \\ \Rightarrow 3\pi - x = 10 \end{cases}$$

$$\text{and } y = \cos^{-1}(\cos 10) \quad \begin{cases} 3\pi < 10 < 4\pi \\ \Rightarrow 4\pi - x = 10 \end{cases}$$

$$\Rightarrow y = 4\pi - 10$$

$$\therefore y - x = (4\pi - 10) - (3\pi - 10) = \pi$$

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Q3

$$(1) \cot \left(\sum_{n=1}^{19} \cot^{-1} \left(1 + \sum_{p=1}^n 2p \right) \right)$$

$$= \cot \left(\sum_{n=1}^{19} \cot^{-1} (1 + n(n+1)) \right)$$

$$= \cot \left(\sum_{n=1}^{19} \tan^{-1} \left(\frac{(n+1) - n}{1 + (n+1)n} \right) \right)$$

$$\left[\cot^{-1} x = \tan^{-1} \left(\frac{1}{x} \right) : \text{for } x > 0 \right]$$

$$= \cot \left(\sum_{n=1}^{19} (\tan^{-1}(n+1) - \tan^{-1} n) \right)$$

$$= \cot(\tan^{-1} 20 - \tan^{-1} 1)$$

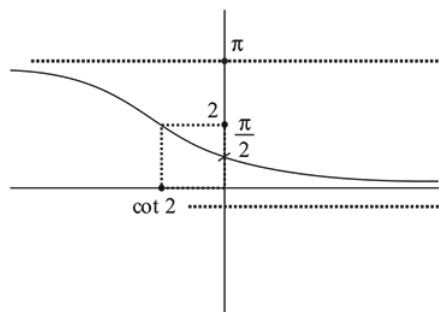
$$= \cot \left(\tan^{-1} \left(\frac{20-1}{1+20 \times 1} \right) \right)$$

$$= \cot \left(\tan^{-1} \left(\frac{19}{21} \right) \right) = \cot \cot^{-1} \left(\frac{21}{19} \right) = \frac{21}{19}$$

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Q4

(2)



$$(\cot^{-1} x)^2 - 7(\cot^{-1} x) + 10 > 0$$

$$(\cot^{-1} x - 5)(\cot^{-1} x - 2) > 0$$

$$\cot^{-1} x \in (-\infty, 2) \cup (5, \infty) \quad \dots(1)$$

But $\cot^{-1} x$ lies in $(0, \pi)$

Now, from equation (1)

$$\cot^{-1} x \in (0, 2)$$

Now, it is clear from the graph

$$x \in (\cot 2, \infty)$$

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Q5

(3) Consider, $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$

$$\Rightarrow \tan^{-1}\left(\frac{5x}{1-6x^2}\right) = \frac{\pi}{4}$$

$$\Rightarrow \frac{5x}{1-6x^2} = 1 \Rightarrow 5x = 1 - 6x^2$$

$$\Rightarrow 6x^2 + 5x - 1 = 0$$

$$\Rightarrow (6x - 1)(x + 1) = 0$$

$$\Rightarrow x = \frac{1}{6} \text{ (as } x \geq 0)$$

Therefore, A is a singleton set.

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Q6

(4) $\because \cos \alpha = \frac{3}{5}$, then $\sin \alpha = \frac{4}{5}$

$$\Rightarrow \tan \alpha = \frac{4}{3}$$

and $\tan \beta = \frac{1}{3}$

$$\because \tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \cdot \tan \beta}$$

$$= \frac{\frac{4}{3} - \frac{1}{3}}{1 + \frac{4}{3} \cdot \frac{1}{3}} = \frac{\frac{3}{3}}{\frac{13}{9}} = \frac{9}{13}$$

$$\therefore \alpha - \beta = \tan^{-1}\left(\frac{9}{13}\right) = \sin^{-1}\left(\frac{9}{5\sqrt{10}}\right)$$

$$= \cos^{-1}\left(\frac{13}{5\sqrt{10}}\right)$$

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Q7

(1) Given, $\cos^{-1} x - \cos^{-1} \frac{y}{2} = \alpha$

$$\Rightarrow \cos^{-1}\left(\frac{xy}{2} + \sqrt{1-x^2} \cdot \sqrt{1-\frac{y^2}{4}}\right) = \alpha$$

$$\Rightarrow \frac{xy}{2} + \frac{\sqrt{1-x^2} \sqrt{4-y^2}}{2} = \cos \theta$$

$$\Rightarrow xy + \sqrt{1-x^2} \sqrt{4-y^2} = 2 \cos \alpha$$

$$\Rightarrow (xy - 2 \cos \alpha)^2 = (1-x^2)(4-y^2)$$

$$\Rightarrow x^2 y^2 + 4 \cos^2 \alpha - 4xy \cos \alpha = 4 - y^2 - 4x^2 + x^2 y^2$$

$$\Rightarrow 4x^2 - 4xy \cos \alpha + y^2 = 4 \sin^2 \alpha$$

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Q8

$$(2) \quad -\sin^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{12}{13}\right) = -\sin^{-1}\left(\frac{3}{5} \times \frac{5}{13} - \frac{12}{13} \times \frac{4}{5}\right)$$

$$(\because xy \geq 0 \text{ and } x^2 + y^2 \leq 1)$$

$$\left[\because \sin^{-1} x - \sin^{-1} y = \sin^{-1} \left\{ x\sqrt{1-y^2} - y\sqrt{1-x^2} \right\} \right]$$

$$= \sin^{-1}\left(\frac{-33}{65}\right) = \sin^{-1}\left(\frac{33}{65}\right)$$

$$= \cos^{-1}\left(\frac{56}{65}\right) = \frac{\pi}{2} - \sin^{-1}\left(\frac{56}{65}\right)$$

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